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Traditional vs naturalised design: a comparison of affordances and physical activity in two preschool playscapes

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ABSTRACT

This study employed a range of observation methods engaging with the concept of affordances to create a qualitative snapshot of children's movement behaviour in two playscapes. Comparison of children's movement behaviour in the playscapes generated the research question of whether a highly naturalised playscape provides more varied opportunities for physical activity than traditionally designed playscapes with standard equipment. Based on observations of children's activity in the playscapes there appeared to be differences in manipulative activity in the naturalised space. Further analysis using behaviour mapping indicated the naturalised space may offer potentially varied opportunities for developing children's sense of balance and increasing physical activity. Children made wider use of the naturalised space and resources, whereas activity in the traditional space was more likely confined to standard areas and equipment. The findings suggest naturalised design potentially provides equivalent or more varied opportunities for preschooler physical activity than traditional designs and standardised equipment.

KEYWORDS

Preschool; physical activity; movement; playscape design; affordances; naturalised playscapes; traditional playscapes

1. Introduction

Quality outdoor playscapes within early childhood programmes are designed to provide opportunities for young children to engage in a variety of physical and movement activities, to support different aspects of their physical development. These include fitness, endurance, agility, coordination, and fine motor control (Woolley & Lowe, 2013). Traditionally, standard forms of specifically designed and often fixed and/or single use equipment have been regarded as essential fittings for outdoor playscapes for young children, such as climbing frames, slides, balance beams, swings, sliding poles, and climbing/sliding forts (Woolley & Lowe, 2013). Each type of equipment is typically designed to support one or more specific forms of activity, such as climbing, balancing, sliding. Older forms of the traditional playground often included areas of hard surfacing for activities such as running and the use of wheeled vehicles. Newer versions, what Woolley (2008) termed 'Kit, Carpet and Fence' (KFC kit) playgrounds with a 'kit' of pre-fabricated equipment of the equipment-based playgrounds, will have safety surfacing such as rubber, with limited provision of other types of hard or soft surfacing (Brunelle, Herrington, Coghlan, & Brussoni, 2016; Woolley, 2008).

A recent development in the design of outdoor spaces for young children has been the provision of spaces that more resemble natural environments incorporating features such as trees and shrubs, rocks, dirt, loose parts, uneven surfaces, and inclines (Refshauge, Stigsdotter, Lamm, & Thorleifsdottir, 2015; Woolley & Lowe, 2013). These more naturalised designs appear to increase the level and variety of young children's physical activity, and contribute to the enhancement of their motor development (Gill, 2014). For example, researchers have found that features such as uneven ground, slopes, rocks, trees and vegetation appear to provide the necessary challenges and demands that support both higher levels of activity and the development of motor coordination and agility (Boldemann et al., 2006; Fjørtoft, 2001; Fjørtoft & Sageie, 2000; Grahn, Martensson, Linblad, Nilsson, & Ekeman, 1997; Woolley & Lowe, 2013).

On the other hand, Storli and Hagen (2010) contend that there is still much to find out about whether and how different environmental features and levels of naturalisation may support different rates or types of physical activity in children. For example, in a study using accelerometers, they found no significant difference in levels of physically active play between a traditional playground and natural environment (Storli & Hagen, 2010). Nicaise, Kahan, and Sallis (2011) found that design elements such as large open spaces including grass fields, and equipment including activity-genic portables, gross motor manipulatives and riding vehicles, can foster moderate to vigorous physical activity (MVPA) in pre-schoolers (Nicaise et al., 2011). Cosco, Moore, and Islam (2010) found higher levels of moderate and vigorous motor activity occurred on hard surfaced, curved pathways that fostered running and use of wheeled toys.

This range of findings raises the question of whether traditionally equipped playscapes and naturalised playscapes offer equivalent or differentiated support for children's physical development (Refshauge et al., 2015). There are also questions around the measurement of different aspects of physical activity and development in outdoor landscapes. Luchs and Fikus (2018) note a variety of measures in past studies, each with particular limitations, leading to a diversity of findings warranting further research. For example, Fjørtoft's (2001) study looked at child outcomes on a variety of tests of agility and co-ordination, while Nicaise et al. (2011) and Kahan, Nicaise, and Reuben (2013) considered preschool outdoor physical activity via accelerometry and direct observation. Cosco et al. (2010, p. 515) tracked 'gender, behaviour setting type, physical attributes, and physical activity level using CARS' (modified Child Activity Rating Scale) via behaviour mapping. Other researchers (Storli & Hagen, 2010) have used robust quantitative measures of physical activity levels (accelerometry) that may not account for greater diversity or complexity of movement behaviours.

It may be that traditional and naturalised playscapes each support different aspects of children's movement development. For example, do traditional spaces with hard surfaces encourage more vigorous activity such as bike riding? Are the uneven surfaces and challenge of obstacles in more naturalised spaces more likely to support the development of children's agility and coordination?

It is understood that more naturalised spaces may enhance children's manipulative play. Nicholson (1972) first linked the number of 'loose parts' of the sort found in naturalised environments (sticks, stones, leaves, sand) to environmental complexity, and argued the 'number and kind of variables' available to children influenced creativity, inventiveness and discovery (p. 6). Research has revealed manipulative environments that offer natural loose parts provide enhanced opportunities for manipulative play and engagement in problem solving, creativity, and social interaction (Zamani & Moore, 2013).

In comparison with more traditional playscapes characterised by standard equipment or the newer KFC kit designs (Woolley, 2008), natural environments have been described as providing children with more complex, evolving, and interesting spaces that promote motor activity, and afford natural obstacles and challenges that children cope with and eventually overcome (Fjørtoft, 2001; Herrington & Lesmeister, 2007; Refshauge et al., 2015). However, when appraising the contextual evidence for design influences upon young children's active play behaviour in early childhood outdoor landscapes, it is important to note that research studies that support and

explains these associations have not necessarily been conducted in comparable settings to each other, or to the setting under investigation in this study. Significant studies, for example, have focused on children between the ages of one to 12 years conducted in public playground settings (Refshauge et al., 2015; Woolley & Lowe, 2013) and schools (Dyment & Bell, 2007) rather than specific early childhood settings. Conversely, Refshauge et al. (2015, p. 243) noted potential comparability issues, where evidence from childcare research was used to inform their study focused upon public playgrounds.

2. Concept of affordances

In seeking to understand how traditional and natural outdoor playscapes may differentially influence children's physical activity and development, Gibson's (1979) theory of *affordances* provides a useful framework (Gibson, 1979). A functional approach to understanding children's relation to their environment has the potential to explain how different features and aspects of the physical environment 'afford' children specific experiences or opportunities for action (Heft, 1988; Lerstrup & Konijnendijk van Den Bosch, 2017). Affordances are essentially 'relational', being a product of the relationship between the environmental feature and the individual (Heft, 1989). Encompassing characteristics of both the environment and the person, they are unique in character. Children will respond individually to environmental features according to age, stage of development, motivation, past experiences (Little & Sweller, 2015). For example, a tree that affords opportunities to hide for a younger child might afford climbing, hanging, or balancing on for an older more capable child.

Affordances can also be considered as either *potential* or *actualised* (Kytta, 2004; Refshauge et al., 2015). As Heft, (1989) notes: 'Affordances constitute the ecological resources of the environment that *may* (emphasis added) be utilised by the individual. Whether particular affordances are utilised ...will depend on intentional processes of the perceiver.' (1989, p. 10). Designers and other adults may provide what they see as potential affordances, but whether they become actualised depends on children's responses. Because adults tend to perceive play spaces in terms of forms such as conventional play equipment and materials, while children tend to see possibilities for actions or experiences (Heft, 1988), children may not perceive potential affordances in the same way as adults, and may even perceive affordances that designers haven't thought of (Refshauge et al., 2015).

For example, Morrissey, Scott, & Wishart, (2015) looked at toddler responses to a redesign of their outdoor play space. While children responded as expected to many of the potential affordances that the designers built into the design, there were other affordances actualised by the children that designers had not foreseen. These included children's use of garden edgings for balancing (whereas a commercially produced balance beam was used only for sitting), and use of a bridge on a platform for climbing.

So far, research is limited on how children perceive affordances, and how their perceptions may or may not align with designer expectations. Observations of children's activity in an environment, and their use of environmental features, provides information about actualisable affordances in a setting. As Kytta states: 'Actualised affordances are revealed through the actions of the individual, or through self-report' (2004, p. 181). For designers and researchers, evidence of perceived and actualised affordances, both expected and unexpected, can be regarded as evidence of design effectiveness (Refshauge et al., 2015).

3. The study

Using an affordances lens, this study compared children's physical activity in a traditional outdoor playscape, and a new naturalised space within the same preschool centre located in a regional town in Victoria, Australia. A pivotal question for the designers and researchers was whether the

naturalised design of the new space could provide equivalent actualisable affordances for different types of physical activity to those provided by the more traditional playspace, with its conventional equipment and resources.

The design of the new area aimed to provide maximum affordances for play and physical activity through naturalised and embedded features. For example, a fairly steep rise, or 'hill', was terraced on one side, with steps and embedded slides intended to provide affordances for climbing, sliding and balancing. On the other side, children could travel up and down the slope through walking or running. Apart from the embedded slides, there was no traditional equipment. The two playscapes were located on different sides of the building, each extending to the back of the building where a connecting gate was sometimes opened to allow free-flow of children between the two spaces.

The centre participated in a larger development and research programme involving a collaboration between service providers, a nursery/landscape company, and university researchers. The larger project's aim was the development and evaluation of innovative naturalised outdoor play spaces. It drew on affordances principles and what has been called the *evidence-based landscape architecture approach*, based on design principles and practice, stakeholder information, and research evidence (Brown & Corry, 2011; Refshauge et al., 2015).

4. Method

4.1. The two playscapes

4.1.1. The traditional playscape

The original outdoor space at the preschool was a traditional design, with standard fabricated equipment and structures providing typical affordances for physical activity. These included: a fort incorporating slide, ladders and swing-bridge; climbing frames; sandpit; large wooden ship; synthetic surfaced open area for bikes and sports; cubby house; teepee; wooden stage; swings; and a small rowing boat (see Figure 1). This space also contained natural features including a grassed area, vegetable garden, and well-established trees and shrubs (see Figures 1–4). Such features can provide affordances for running, and loose parts for manipulation. This space can be characterised as 'traditional' in that, the overarching (design) focus is on the

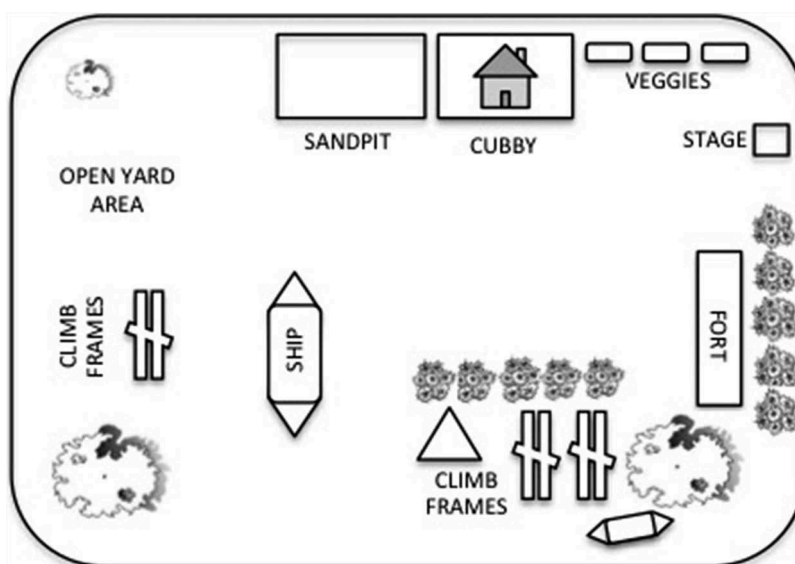


Figure 1. Diagrammatic layout and overview of the 'Traditional Space'.



Figure 2. Traditional outdoor space southeast diagonal view including blue boat and fort fixtures.

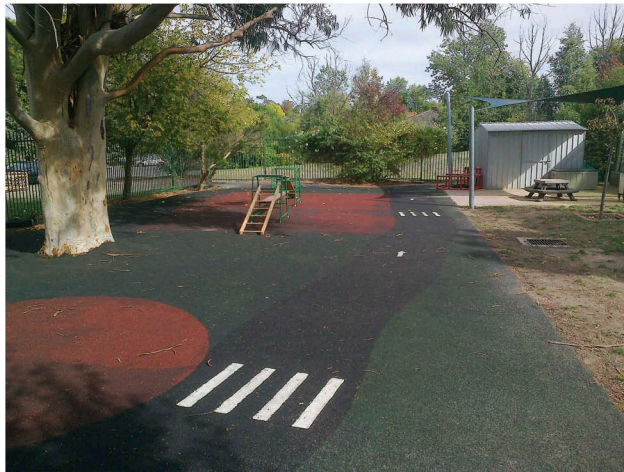


Figure 3. Traditional outdoor space facing south, open yard area with synthetic surface and movable A-frame and balance beam in view.

provision of a traditionally organised variety of spaces, fixtures and structures to support children's play and development, albeit with some natural features that shift it a little on the continuum towards Woolley and Lowe's 'composite' type design (Lerstrup & Konijnendijk van Den Bosch, 2017; Woolley & Lowe, 2013).

4.1.2. *The new naturalised playscape*

The new playscape can be easily classified as being at the 'nature' end of Woolley and Lowe's continuum (Woolley & Lowe, 2013) and similar to what Herrington and Lesmeister (2007) characterised as an 'organic' play space (p. 76). A design goal was to replace traditional fabricated equipment and built resources where possible with natural features (or integrate them with natural features), drawing on the concept of affordances. For example, the playscape was designed with a series of terraces, edgings, inclines, logs, and rocks designed to provide potential affordances for physical activities such as climbing, balancing and negotiating inclines (see Figures 5–9). Play areas were loosely defined through natural materials such as rocks, logs and bushes (see Figures 8–10).



Figure 4. Traditional outdoor space southwest diagonal view with movable climbing fixtures.

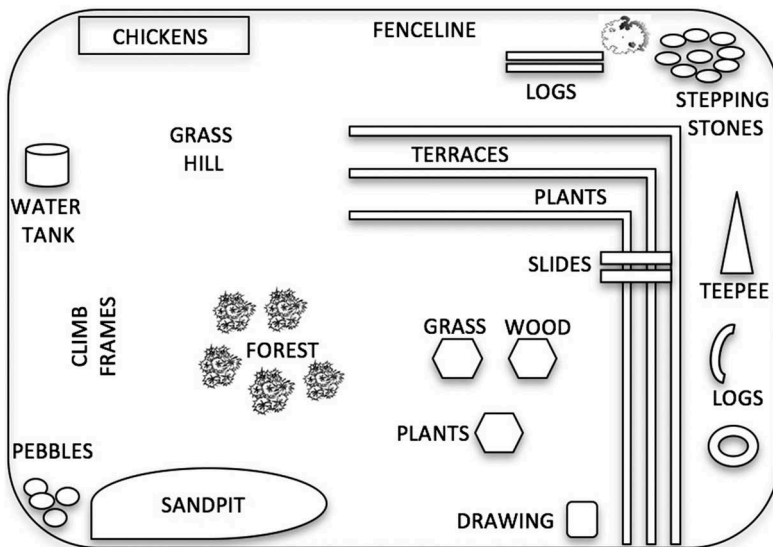


Figure 5. Diagrammatic layout and overview of the 'Naturalised playscape'.

Natural features included: terraced gardens with fruit trees; herb garden and small plants; logs; stepping-stones; log enclosure; small tree forest; sandpit with pebbles and medium-size rocks. Stand-alone or fabricated structures and elements were few. Surfaces consisted predominantly of natural materials such as grass, soft fall and bare soil, providing largely uneven and varying terrain, compared to the predominantly level surfaces in the traditional playspace. A distinguishing feature was a steep embankment taking up half the area. There were a variety of 'pathways' or access points for children to move up and down the embankment area including the grassed slope, the embedded slides, and a range of steps and terraces. Two slides and a rope climbing net were embedded in the embankment. This area provided affordances for vigorous physical activity such as: log rolling down the embankment; running up and down the slope/hill and terraces; sliding; balancing; negotiating uneven surfaces, inclines, and 'obstacles' such as bushes, rocks and trees. In terms of potential affordances, the new naturalised space offered greater variety and challenge.



Figure 6. Naturalised outdoor space elevated northeast diagonal view including wood, grass and plant areas with terraces and embedded slides.



Figure 7. Naturalised outdoor space facing south with view of forest and grass hill areas.
Note: photo by Fleming's Nurseries Pty Ltd. Reproduced with permission.

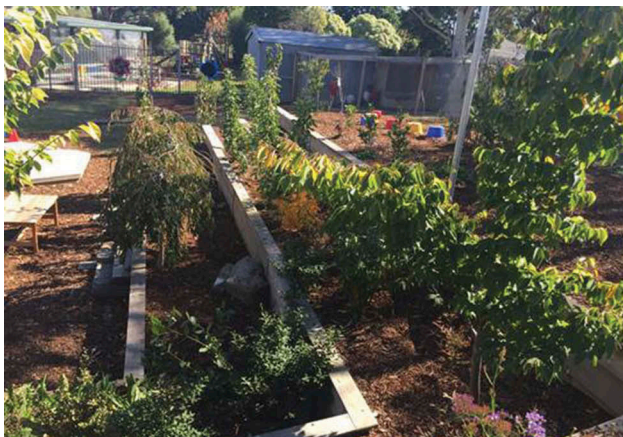


Figure 8. Naturalised outdoor space side view of terraces, trees and shrubs: view from above embankment.

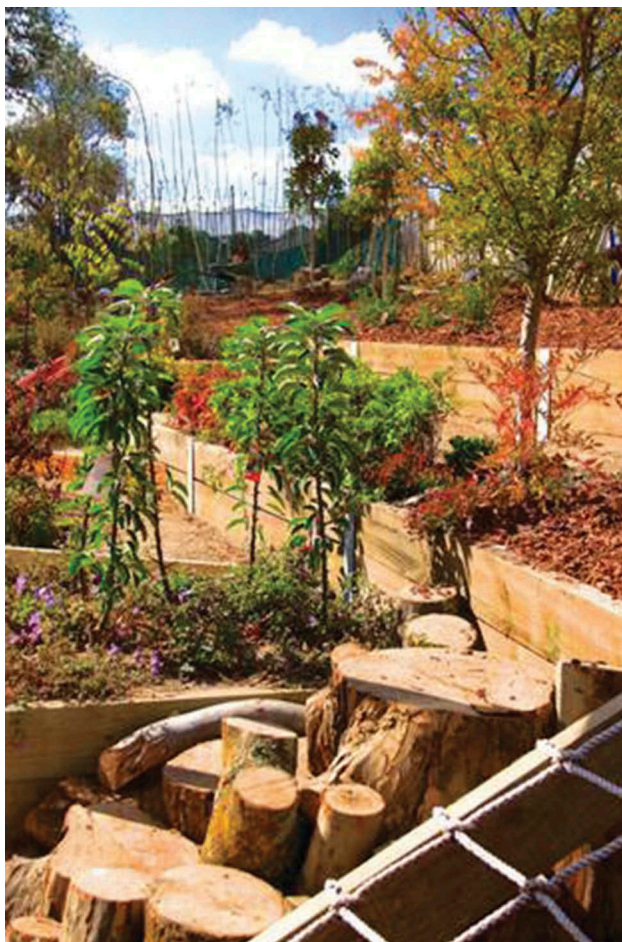


Figure 9. Naturalised outdoor space view from below of terraces, ladder, wood pieces, trees, and shrubs: view from below embankment.

Note: photo by Flemings Nurseries (2016). Reproduced with permission.



Figure 10. Naturalised outdoor space at top of embankment with natural structures and fixtures in view.

It should be noted that this study was conducted in the year following the establishment of the new naturalised space. As [Figures 5–9](#) show, the trees and other vegetation had not reached full growth, and so may not have been manifesting their full impact as natural design features. As an example, in the ‘forest’ area the closely planted trees were still short, with branches growing at children’s height. This limited play in this area at this time, and the full potential of children being able to play under a tree canopy had not yet been realised.

4.2. Participants

Study participants were 4–5-year-old children nearing the end of their preschool year. They attended the preschool for an average of 15 h per week, in three groups of approximately 25 each. Children from these three groups were observed in active play in their respective outdoor playspaces in varied numbers up to a maximum of 25 in any given observation period. According to recent 2015 Australian Early Development Census data, children in this preschool’s community showed low vulnerability in their physical, language, social and communication development, but higher developmental vulnerability in the area of emotional development (14%) compared to the Victorian state average (8%) (Commonwealth of Australia, 2015).

The research was part of a larger study that was conducted according to the ethical requirements of the University and the relevant state government education department. Parental consent was obtained for the participation of all children in the project, and measures were taken to ensure the confidentiality of the centre and all participants.

4.3. Data collection

For the purposes of behaviour mapping (Cosco et al., 2010; Kutnick, Brighi, & Colwell, 2016) a time-sampling observation tool was developed based on typologies and qualitative characteristics of movement and physical activity (see [Appendix A](#)) adapted from earlier studies by the authors (Morrissey et al., 2015; Rouse, Wishart, Morrissey, & Scott, 2014). Observations were conducted between the hours of 10:30–15:30 on weekdays during preschool sessions. The two playscapes were each divided into zones, each with different affordances, and children’s activity in each area was observed in a rotating cycle of three-minute time samples. This enabled researchers to collect data on overall physical activity within each space, and in relation to specific affordances. The preschool routines meant that the numbers of children outside at any one time varied. Hence, summing of total numbers of children and specific breakdown of numbers of boys and girls in each observation was not undertaken.

There was a focus on the type and quality of children’s movement, as past research has suggested naturalised outdoor spaces make a unique contribution to the intensity, range, quality and motivation of children’s physical activity (Fjørtoft & Sageie, 2000; Gill, 2014). For each observation, the tool also noted:

- Number of boys and girls (no further count of children was taken);
- Presence of educators
- whether play was solitary or group;
- location and general contextual information.

[Table 1](#) summarises the major coding categories for children’s movement behaviour. *Locomotor* refers to children’s fundamental movements such as walking, skipping, running and jumping. *Translocation* refers to children transiting through or across a stipulated area of observation to another part of the outdoor space. *Manipulative* refers to children’s manipulative movement play with manufactured objects such as balls or spades and natural objects or materials such as sticks, pebbles, and rocks. *Balance* refers to instances where children actively engage with balancing (stabilising movements) or on-off balance movements such as balancing on rocks, logs or traditional play equipment. *Sedentary*, *Moderate* and

Table 1. Physical activity-movement data coding explanatory table.

Coding Categories	Description/Explanation	Reference Points
Type of Movement and FMS ● Locomotor ● Translocation ● Manipulative ● Balance (Stable/Unstable)	Type of movement and Fundamental Movement Skill using Locomotor, Stabilising & Manipulative movement nomenclature (see Reference Points).	(Gallahue & Cleland Donnelly, 2003; Gallahue et al., 2012)
Intensity of Movement ● Sedentary ● Moderate ● Vigorous	Levels of physical activity including; sedentary behaviours (stationary) and moderate through to vigorous intensity physical activity (MVPA). Modelled after pre-schooler physical activity observational studies (see Reference Points).	(Brown et al., 2006; Kahan et al., 2013; Nicaise et al., 2011; Truelove et al., 2018)

Vigorous refers to a continuum of physical activity levels for example: crouching or sitting (Sedentary), to walking (Moderate) to running or sprinting (Vigorous).

To establish inter-coder agreement with the tool, researchers at another site conducted two sets of 40 trial observations over a four-week period prior to the current study (Mukherji & Albon, 2010). Inter-coder agreement was assessed across 27 variables and ranged from 77.5% to 100% (Mean 92%, Median 95%). The lowest agreement was for the variable rating the intensity of play as ‘Moderate’ where there was disagreement in 9 of the 40 observations (77.5% agreement). Five of the variables had disagreement in 7 (82.5% agreement) to 5 (87.5% agreement) of the observations, and the remaining 21 variables had disagreement in 4 or fewer (>90% agreement) of the observations.

Initially two sets of 10 observations were conducted in the outdoor spaces during normal preschool session times by two researchers in order to: establish the observation areas for each respective outdoor space; re-confirm internal validity of coding categories; and, re-check inter-coder agreement so observations could continue with one researcher. Based on the initial round of observations, the researchers excluded from the time-sampling several areas where children rarely played. One researcher continued with 40 observations in the naturalised space and 42 observations in the traditional space from April to June in the year of the study.

4.4. Analysis

4.4.1. Behaviour mapping analyses

According to Kutnick et al. (2016), mapping enables researchers to ‘ascertain characteristic actions of a setting or comparisons between settings’ (p. 271). Behaviour mapping tracked the incidence of different categories of movement across different areas of the two playscapes, to investigate if different categories of movement were more likely to occur in specific areas or in relation to specific features (Cosco et al., 2010).

The mapping also had the potential to provide insight into the *quality* of the movement activities that children engaged in. For example, running across uneven, rough surfaces may entail greater challenge and more demands on children’s agility and coordination as compared to running across a smooth level surface (Fjørtoft, 2004). It was also expected that the mapping data could identify areas or features where affordances for different types of movement were more likely to be actualised (Cosco et al., 2010; Morrissey et al., 2015). Complementing the Behaviour mapping were a series of analyses comparing the traditional playscape and the naturalised playscape (see 4.1) using the outcome variables in continuous and binary format (detailed methods are described in Supplemental Material).

5. Results

5.1. Activity level comparisons

Manipulative play was observed in 34 of the 40 observations in the naturalised space (85%) compared to 19 of 42 observations in the traditional space (45.2%). Further analysis coded

manipulation as involving either natural or manufactured objects and materials (Gallahue & Donnelly, 2003; Gallahue, Ozmun, & Goodway, 2012). This showed a higher rate of manipulation of natural materials in the naturalised playscape where 28 of 34 observations (82.4%) involved manipulation of natural materials compared to 5 out of 19 (26.3%) in the traditional playscape.

5.1.1. Gender

As noted in 3.3, gender as an observation category only recorded presence (regardless of whether or not they were engaged in physical activity) of boys and girls in each observation not actual numbers by gender. Data analysis showed the following:

New Yard—Boys: Yes = 37 and No = 3 (Total number of observations 40).

New Yard—Girls: Yes = 25 No = 15 (Total number of observations 40).

Old Yard—Boys: Yes = 31 No = 10 (Total number of observations 41).

Old Yard—Girls: Yes = 39 No = 2 (Total number of observations 41).

Since the total number of boys and girls in each individual observation was not recorded, the researchers decided not to pursue making further inferences based on this data.

5.1.2. Educator presence

Data on the presence/absence of educators showed:

- New Yard: Absent = 17 out of 40 observations. Present = 10 out of 40 observations. Sometimes = 13 out of 40 observations
- Old Yard: Absent = 29 out of 41 observations. Present = 3 out of 41 observations. Sometimes = 9 out of 41 observations.

Once again, the researchers decided not to pursue further inferences from the data.

5.2. Mapping of activity across the two playscapes

Based on the behaviour mapping, the researchers created a series of schematic maps (Figures 11–16) with each map showing the locations of the time-sampling observations of the different movement categories. For instance, Figures 11 and 12 illustrate the different distributions of manipulation of natural and manufactured objects and materials across each playscape. Analysis was conducted to see if the mappings (schematisation) could yield any insights into the actualisation of affordances in each playscape. Subsequently, the researchers felt that the mapping data for two of the categories, *manipulative play* and *intensity of movement*, and one of the Types of Play sub-categories—*balance*—provided opportunities for further interpretation and insight (see Figures 11–16).

5.2.1. Manipulative play

Figure 11 shows that most manipulative play in the traditional setting occurred in the sandpit or the open yard area. Manipulative play in the sandpit involved typical affordances provided by manufactured toys such as buckets, shovels, spoons, pots, trucks, rakes, wheel barrels, and dinosaur figures. In the open yard area, it also involved manufactured objects such as hula-hoops, tricycles, balls and wheelbarrows. There were few recorded instances of actualisation of natural affordances for manipulation in the traditional space. One involved children coming out to play and in excitement throwing leaves up in the air. Other examples included carrying sand in and out of the sand pit, and the use of a stick with a wheelbarrow.

This contrasts with observations in the naturalised space, where manipulative play involved frequent actualisation of affordances provided by natural materials and loose parts such as leaves, bark, rocks, sticks, and sand. Manipulation of natural materials was also widely dispersed across this space (Figure 12). There was some manipulative play with manufactured objects, such as buckets, bowls and cans, mostly in the sandpit and the areas close to water/mud play.

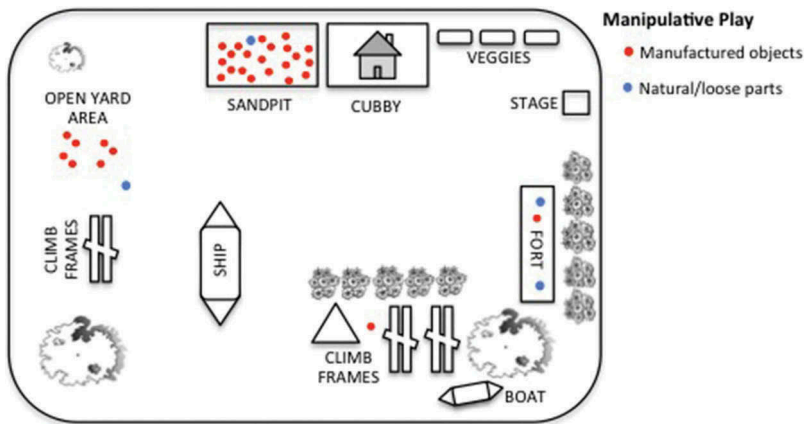


Figure 11. Instances and locations of manipulative play in the traditional playscape.

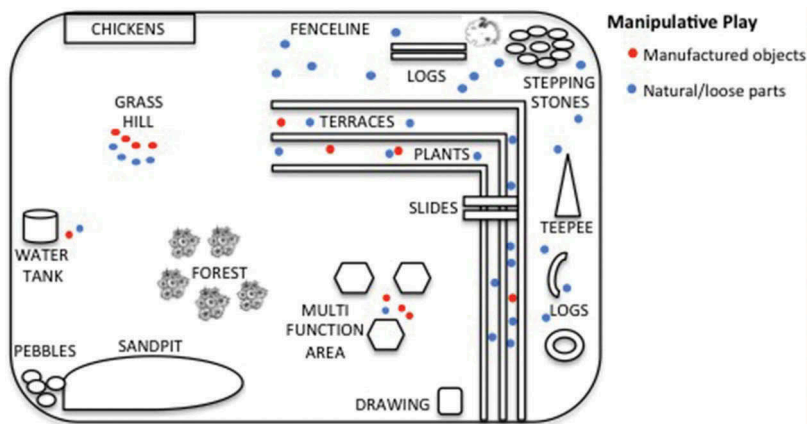


Figure 12. Instances and locations of manipulative play in the naturalised playscape.

5.2.2. *Balancing*

Figure 13 shows that in the traditional playscape most balancing activity occurred in the open yard area, with a few instances around the A-Frame, blue boat and fort. A lot of the balancing in the open area happened on a wooden balance beam supported by two A-frames. Thus, the mapping shows balancing in the traditional space mostly involving affordances provided by fabricated structures, and on level and synthetic surfaces.

By contrast, Figure 14 shows that balancing activity in the naturalised playscape was more widely distributed and involved different features including rocks, logs and stumps, and the edgings on the sandpit and the terraces. Balancing in the naturalised space was also occurring in situations involving heights and inclines (the slope of the hill and on the terraces). In contrast to the traditional playscape, mapping shows the balancing in the naturalised playscape largely involved the extra challenges of actualising affordances provided by a variety of uneven and irregular natural surfaces and features, as well as height and slope.

5.2.3. *Intensity of movement*

Figures 15 and 16 show the distribution of the different levels of intensity of movement in each of the two playscapes. In the traditional space (Figure 15), sedentary activity was

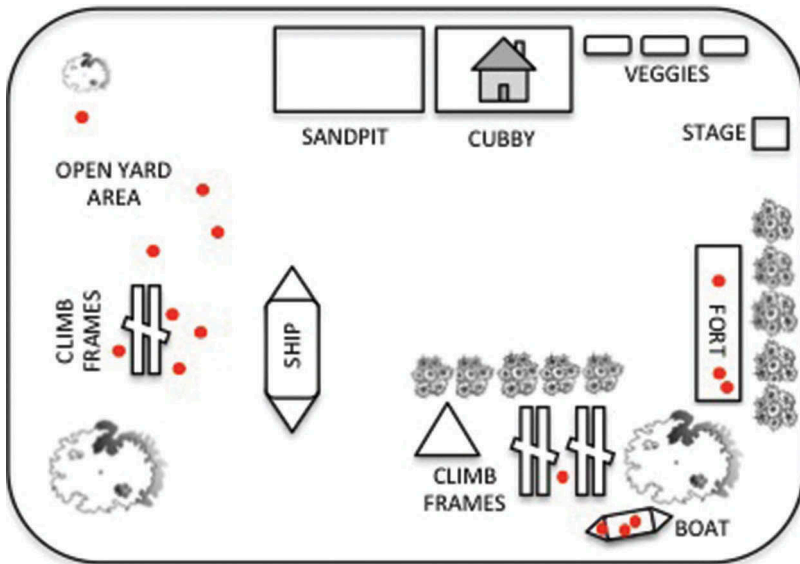


Figure 13. Type of movement: Instances and locations of balance in the traditional playscape.

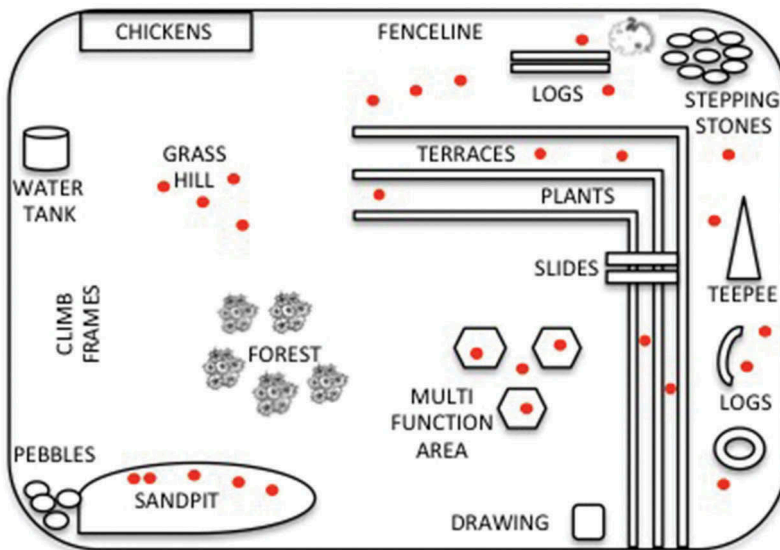


Figure 14. Type of movement: Instances and locations of balance in the naturalised playscape.

distributed fairly evenly across the different areas, apart from the blue boat. Moderate intensity activity was also evenly spread across the whole space. Vigorous activity was most frequent in the open yard area, on the A-Frames, and with a couple of instances on the boat and one occurring on the fort. Considering how different spaces and equipment afforded opportunities for different intensities of movement, the mapping data indicates that the open yard area and the A-Frames provided affordances for all three levels of movement intensity, while the sandpit and the fort yielded only sedentary and moderate activity, with only one instance of vigorous activity in the fort.

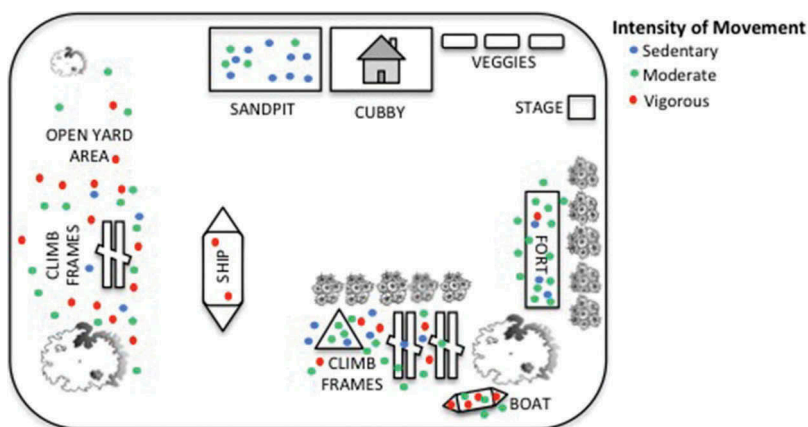


Figure 15. Intensity of movement: Instances and locations of sedentary, moderate, and vigorous movement in the traditional playscape.

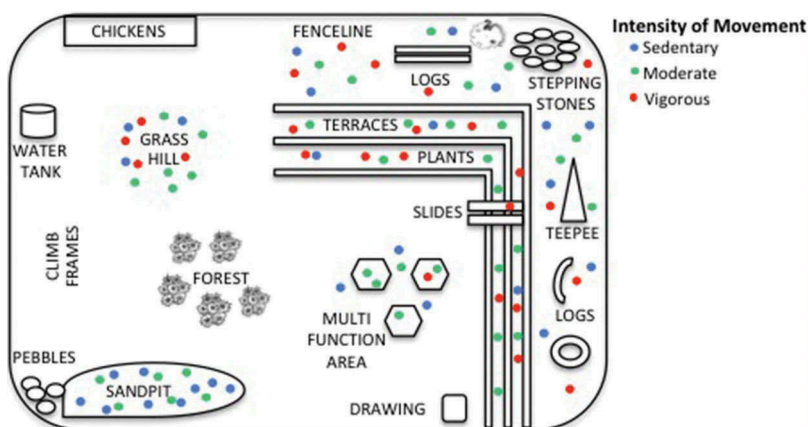


Figure 16. Intensity of movement: Instances and locations of sedentary, moderate, and vigorous movement in the naturalised playscape.

In the naturalised playscape, Figure 16 shows sedentary and moderate intensity activity evenly distributed across the space. Vigorous intensity activity occurred on the hill area, the terraces, and fence-line areas behind. As with the traditional playscape, there was no vigorous activity in the sandpit area. Compared to the traditional space, vigorous activity in the naturalised setting was more widespread, and occurred with a more diverse range of spaces and resources. As with balance activities, the naturalised space provided qualitatively different opportunities for moderate and vigorous movement activity, involving actualisation of more demanding affordances including uneven surfaces, inclines, narrow edgings, differing levels and terraces, and an abundance of natural and fabricated 'obstacles' to be negotiated (bushes, trees, logs, rocks, steps, maze).

In summary, analysis showed that compared to the traditional playscape, children in the naturalised setting were significantly more likely to engage in manipulative activity, and to do this with natural materials. Apart from manipulation, the rates of different types of movement, and levels of movement intensity were similar across both playscapes. Mapping data, however, indicated qualitative differences between spaces in balance and in moderate and vigorous movement activities, with children actualising more varied and challenging affordances available in the naturalised space.

6. Discussion

One of the design purposes of the naturalised playscape was to replace traditional fabricated and fixed-use equipment and resources as much as possible with natural features and resources. This was based on the premise that carefully designed natural spaces and features can provide similar affordances to traditional fabricated resources and equipment for various forms of physical activity (Lerstrup & Konijnendijk van Den Bosch, 2017). It was hoped that the new, highly naturalised space would support equivalent or perhaps even higher levels and quality of physical movement. Results of data analysis showed that this goal was achieved, with higher frequencies of manipulative activity in the naturalised space, and equivalent rates of the other movement categories in both spaces. In addition, the mapping showed that children were actualising the more challenging affordances for balance and movement available in the naturalised space.

In regard to manipulative activity, the results of this study align with other evidence showing that a naturalised playscape can provide an abundance of affordances for children's interaction with natural materials, and that children are inclined to actualise these affordances (Kytta, 2004). While there were no significant differences between the two playscapes in the frequencies of different types of movement, mapping showed that the qualitative characteristics of the actualised affordances for these activities were different in the two spaces. In the more traditional space, they largely involved standard equipment and level grassed and artificially surfaced areas. Unlike the naturalised space, they did not involve the extra challenges and demands of uneven and unpredictable surfaces, varying heights and inclines, and obstacles to be negotiated. These extra challenges and demands suggest that the naturalised space may have provided more affordances for the development of physical skills such as agility and coordination (Fjørtoft, 2001, 2004), as well as the strength and endurance that can be potentially measured by accelerometers (Storli & Hagen, 2010).

Several researchers have found that either there are no significant differences on basic measures of physical activity such as MVPA between traditional and natural playscapes, or that affordances provided by the hard surfaces of traditional playgrounds can lead to greater levels of such activity (Cosco et al., 2010; Storli & Hagen, 2010). Storli and Hagen propose that measures such as MVPA may end up being lower in natural spaces 'because of the difficulties for the children to move around in the demanding surroundings' (2010, p. 455). Analysis showed that this was not the case in this study, as movement levels were similar across the two spaces, while affordances appeared more demanding in the natural space.

While confirmatory measures of children's strength, coordination and endurance were not used in this study, the findings do suggest that the new naturalised space provided development opportunities in these areas. Much of the moderate and vigorous activity in this space occurred on the 'hill' involving a quite steep incline, and up and down the terraces, likely involving extra demands on children's coordination and aerobic and musculoskeletal fitness (Gray et al., 2015), compared to the level surfaces in the traditional space.

Mapping also showed that in the traditional space, activities such as manipulation, balancing and vigorous movement was more likely to be confined to settings and features traditionally associated with the relevant activity. For example, manipulative activity occurred mostly in the sandpit, and most of the balancing activity in the open area on the wooden plank between two A-frames. In the naturalised space on the other hand, children actualised affordances more widely across the space, utilising a variety of natural features for movement activities.

Informally the researchers observed that children's dramatic play in the new playground often involved active movement across the space, with children integrating the terrain and features into their play. For example, one group of children were observed pretending to be horses galloping across an imaginary landscape and 'eating' leaves on trees. This aligns with other findings showing greater utilisation of spaces in natural settings, and that children are more likely to be physically active and move around the space in their play (Kuh, Ponte, & Chau, 2013; Morrissey, Scott, &

Rahimi, 2017). Researchers have suggested that the open-endedness and connectivity of naturalised play spaces encourage greater use of the space and integration of movement and agency within a space compared to traditional spaces dominated by equipment and zoning suggesting narrowly defined uses (Herrington & Lesmeister, 2007; Morrissey et al., 2017).

Overall, the findings show how the use of mapping and an affordance lens can inform evaluation of design effectiveness, aligning with other findings on playspace design (Cosco et al., 2010; Morrissey et al., 2015; Refshauge et al., 2015). It would be useful for future research to conduct larger-scale and longitudinal studies with greater control of variables. Gender as a variable in outdoor play and affordance-focused studies has been highlighted previously (Änggård, 2011; Refshauge et al., 2015; Waller, 2010) and warrants further investigation, as well as the role of early childhood educator engagement (Copeland, Kendeigh, Saelens, Kalkwarf, & Sherman, 2012; McClintic & Petty, 2015; Storli & Hagen, 2010). In addition, the researchers in this study have inferred that the more challenging affordances in the naturalised play space would support the greater development of specific movement skills in children, such as agility, coordination and endurance. Ideally, future studies comparing naturalised and traditional spaces would include specific outcome measures of such skills to test these conclusions.

7. Limitations

The small participant groups and use of one preschool, restricts the generalisability of the findings. There was also limited control of environmental and child participant variables, including timing and duration of observations and fluctuating numbers of children in each outdoor space. The limitations of the observation tool also prevented in-depth investigation of gender and educator engagement effects. In addition, both playscapes contained natural and fabricated elements, preventing a clear-cut comparison of the potential effects of each. The presence of what appeared to be greater total numbers of both natural and built elements (from natural materials) across the naturalised outdoor space may also be an influencing factor for the movement/activity levels in the site.

8. Conclusions

There is evidence of children's declining motor skills in many developed countries (Herrington & Lesmeister, 2007; Refshauge et al., 2015). Emerging data in Australia indicate a rising number of children developmentally vulnerable in terms of physical health and wellbeing, including developmental vulnerability for co-ordination, gross and fine motor skills (Australian Government, 2016). Outdoor playscapes are important sites for young children's movement activity and the development of their physical skills. This study shows that naturalised spaces and integrated built design features can potentially provide equivalent or greater affordances for a range of movement activities compared to a space dominated by standard, fixed and defined-use equipment. In addition, naturalised play spaces have been shown to bring a range of other benefits in children's wellbeing, complexity of play, and sense of agency (Dyment & Bell, 2007; Fjørtoft, 2001; Herrington & Lesmeister, 2007; Morrissey et al., 2017; Refshauge et al., 2015). While natural play spaces may require a greater commitment in terms of ongoing maintenance, the evidence is coming in that in terms of supporting children's optimum development, the investment is worthwhile.

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References

- Änggård, E. (2011). Children's gendered and non-gendered play in natural spaces. *Children, Youth & Environments*, 21 (2), 5–33.
- Australian Government, Department of Education and Training. (2016). *Australian early development census*. Retrieved from <https://www.education.gov.au/australian-early-development-census>
- Boldemann, C., Blennow, H., Dal, H., Martensson, F., Raustorp, A., Yuen, K., & Wester, U. (2006). Impact of preschool environment upon children's physical activity and sun exposure. *Preventive Medicine*, 42(4), 301–308.
- Brown, R. D., & Corry, R. C. (2011). Evidence-based landscape architecture: The maturing of a profession. *Landscape and Urban Planning*, 100(4), 327–329.
- Brown, W., Pfeiffer, K., McIver, K., Dowda, M., Joao, M., Almeida, C., & Pate, R. (2006). Assessing preschool children's physical activity: The observational system for recording physical activity in children-preschool version. *Research Quarterly for Exercise and Sport*, 77(2), 167–176.
- Brunelle, S., Herrington, S., Coghlan, R., & Brussoni, M. (2016). Play worth remembering: Are playgrounds too safe? *Children, Youth and Environments*, 26(1), 17–36.
- Commonwealth of Australia. (2015). *Australian early development census: Data explorer: Table: Percentage of children developmentally vulnerable in 2015*. Retrieved from <http://www.aedc.gov.au/data/data-explorer>
- Copeland, K. A., Kendeigh, C. A., Saelens, B. E., Kalkwarf, H. J., & Sherman, S. N. (2012). Physical activity in child-care centers: Do teachers hold the key to the playground? *Health Education Research*, 27(1), 81–100.
- Cosco, N. G., Moore, R. C., & Islam, M. Z. (2010). Behaviour mapping: A method for linking preschool physical activity and outdoor design. *Medicine and Science in Sports and Exercise*, 42(3), 513–519.
- Dymont, J. E., & Bell, A. C. (2007). Active by design: Promoting physical activity through school ground greening. *Children's Geographies*, 5(4), 463–477.
- Fjørtoft, I. (2001). The natural environment as a playground for children: The impact of outdoor play activities in pre-primary school children. *Early Childhood Education Journal*, 29(2), 111–117.
- Fjørtoft, I. (2004). Landscape as playscape: The effects of natural environments on children's play and motor development. *Children, Youth and Environments*, 14(2), 21–44.
- Fjørtoft, I., & Sageie, J. (2000). The natural environment as a playground for children: Landscape description and analysis of a natural landscape. *Landscape and Urban Planning*, 48(1), 83–97.
- Gallahue, D. L., & Cleland Donnelly, F. (2003). *Developmental physical education for all children*. Champaign, IL: Human Kinetics.
- Gallahue, D. L., Ozmun, J. C., & Goodway, J. (2012). *Understanding motor development: Infants, children, adolescents, adults*. New York, NY: McGraw-Hill Companies, Inc.
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Boston: Houghton Mifflin.
- Gill, T. (2014). The benefits of children's engagement with nature: A systematic literature review. *Children, Youth and Environments*, 24(2), 10–34.
- Grahn, P., Martensson, F., Linblad, B., Nilsson, P., & Ekeman, A. (1997). Outdoors at daycare. *City and Country*, 145, 20–27.
- Gray, C., Gibbons, R., Larouche, R., Sandseter, E. H., Bienenstock, A., Brussoni, M., ... Power, M. (2015). What is the relationship between outdoor time and physical activity, sedentary behaviour, and physical fitness in children? A systematic review. *International Journal of Environmental Research and Public Health*, 12(6), 6455–6474.
- Heft, H. (1988). Affordances of children's environments: A functional approach to environmental description. *Children's Environments Quarterly*, 5(3), 29–37.
- Heft, H. (1989). Affordances and the body: An intentional analysis of Gibson's ecological approach to visual perception. *Journal for the Theory of Social Behaviour*, 19(1), 1–30.
- Herrington, S., & Lesmeister, C. (2007). The design of landscapes at childcare centres: Seven C's. *Landscape Research*, 31 (1), 63–82.

- Kahan, D., Nicaise, V., & Reuben, K. (2013). Convergent validity of four accelerometer cutpoints with direct observation of preschool children's outdoor physical activity. *Research Quarterly for Exercise and Sport*, 84(1), 59–67.
- Kuh, L. P., Ponte, I., & Chau, C. (2013). The impact of a natural playscape installation on young children's play behaviours. *Children, Youth and Environments*, 23(2), 49–77.
- Kutnick, P., Brighi, A., & Colwell, J. (2016). Interactive and socially inclusive pedagogy: A comparison of practitioner- and child-oriented cognitive learning activities involving four-year-old children in preschools in England. *European Early Childhood Research Journal*, 24(2), 265–286.
- Kytta, M. (2004). The extent of children's independent mobility and the number of actualised affordances as criteria for child-friendly environments. *Journal of Environmental Psychology*, 24(2), 179–198.
- Lerstrup, I., & Konijnendijk van Den Bosch, C. (2017). Affordances of outdoor settings for children in preschool: Revisiting Heft's functional taxonomy. *Landscape Research*, 42(1), 47–62.
- Little, H., & Sweller, N. (2015). Affordances for risk-taking and physical activity in Australian early childhood education settings. *Early Childhood Education Journal*, 43(4), 337–345.
- Luchs, A., & Fikus, M. (2018). Differently designed playgrounds and preschooler's physical activity play. *Early Child Development and Care*, 188(3), 281–295.
- McClintic, S., & Petty, K. (2015). Exploring early childhood teachers' beliefs and practices about preschool outdoor play: A qualitative study. *Journal of Early Childhood Teacher Education*, 36(1), 24–43.
- Morrissey, A. M., Scott, C., & Rahimi, M. (2017). A comparison of sociodramatic play processes of preschoolers in a naturalised and a traditional outdoor space. *International Journal of Play*, 6(2), 177–197.
- Morrissey, A. M., Scott, C., & Wishart, L. (2015). Infant and toddler responses to a redesign of their outdoor play space. *Children, Youth and Environments*, 25(1), 29–56.
- Mukherji, P., & Albon, D. (2010). *Research methods in early childhood: An introductory guide*. London: Sage.
- Nicaise, V., Kahan, D., & Sallis, J. F. (2011). Correlates of moderate-to-vigorous physical activity among preschoolers during unstructured outdoor play periods. *Preventive Medicine*, 53(4), 309–315.
- Nicholson, S. (1972). The theory of loose parts: An important principle for design methodology. *Studies in Design Education Craft and Technology*, 4(2), 5–14.
- Refshauge, A. D., Stigsdotter, U. K., Lamm, B., & Thorleifsdottir, K. (2015). Evidence-based playground design: Lessons learned from theory to practice. *Landscape Research*, 40(2), 226–246.
- Rouse, E., Wishart, L., Morrissey, A. M., & Scott, C. (2014). *The harbour outdoors: Investigating how children and educators engage with the outdoor learning space at the harbour family and children's centre* (Unpublished Deakin University report). Melbourne, Victoria: Deakin University.
- Storli, R., & Hagen, T. L. (2010). Affordances in outdoor environments and children's physically active play in pre-school. *European Early Childhood Education Research Journal*, 18(4), 445–456.
- Truelove, S., Bruijns, B. A., Vanderloo, L. M., O'Brien, K. T., Johnson, A. M., & Tucker, P. (2018). Physical activity and sedentary time during childcare outdoor play sessions: A systematic review and meta-analysis. *Preventive Medicine*, 108, 74–85. doi:10.1016/j.ypmed.2017.12.022
- Waller, T. (2010). "Let's throw that big stick in the river": An exploration of gender in the construction of shared narratives around outdoor spaces. *European Early Childhood Education Research Journal*, 18(4), 527–542.
- Woolley, H. (2008). Watch this space! Designing for children's play in public open spaces. *Geography Compass*, 2(2), 495–512.
- Woolley, H., & Lowe, A. (2013). Exploring the relationship between design approach and play value of outdoor play spaces. *Landscape Research*, 38(1), 53–74.
- Zamani, Z., & Moore, R. (2013). *The cognitive play behaviour affordances of natural and manufactured elements within outdoor preschool settings*. Landscape Research, Volume 1. Retrieved from https://works.bepress.com/zahra_zamani/4/

Appendix A

Kindergarten Project—Physical Activity-Movement Data Collection Tool

Researcher	Date	Start Time	End Time
Site	Obs No.	Temperature C°	
Area/Section/Feature		Weather Conditions	
General Impressions of Observation (Notes)			
Gender	Educator		
☐ Boys	☐ Present	☐ Sometimes	
☐ Girls	☐ Absent		
Solitary/Group	Description/Notes		
☐ Solitary			
☐ Group			
Type of Movement and FMS	Description/Notes		
☐ Locomotor			
☐ Translocation			
☐ Manipulative			
☐ Balance (Stable/Unstable)			
Intensity of Movement	Description/Notes		
☐ Sedentary			
☐ Moderate			
☐ Vigorous			
Tempo of Movement			
☐ Slow			
☐ Rapid			
Range of Movement	Description/Notes		
(Effort-Shape: Use of Space and Energy)			
Use of Space (Kinesphere)			
☐ Limited			
☐ Expanded			
Use of Space (Orientation)			
☐ Singular/Simple			
☐ Multiple/Complex			
Effort-Shape			
Yield-Push-Reach-Pull Cycle			
☐ Limited/Simple			
☐ Expanded/Complex			